An Object Detection Framework Based on Relationship Between Objects in an Open Vocabulary Using OWL-VIT And RelTransformer

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Abstract. This paper presents a query-driven framework that extends conventional object detection by jointly identifying objects and their semantic relationships from natural language input. The proposed system integrates OWL-ViT for open-vocabulary object detection, RelTR for visual relationship inference, and a sentence embedding model for query understanding. By leveraging language–vision alignment, the framework enables detection of previously unseen categories and structured relationship reasoning. Experimental evaluation demonstrates that the system achieves an overall recognition accuracy of 42% while effectively generating query-filtered scene graphs. The unified pipeline supports efficient interaction between detection, relational reasoning, and language processing, demonstrating its potential for applications such as intelligent surveillance and context-aware visual analysis.

Keywords: Object Detection, Relationship Detection, Open Vocabulary, OWL-ViT, Rel-Transformer.

1 Introduction

This work addresses the limitation of traditional object detection systems, which typically localize objects without modeling the contextual relationships between them. To enable more comprehensive scene understanding, especially under natural language queries, the study proposes a hybrid framework that jointly performs object detection, relationship reasoning, and query interpretation. The approach integrates OWL-ViT for open-vocabulary detection, RelTR for visual relationship inference, and a language model for processing user queries. Unlike prior methods that rely on individual models or computationally expensive end-to-end solutions, the proposed system is designed to balance flexibility, accuracy, and efficiency. Experimental evaluations on the Visual Genome dataset and a collected dataset, along with ablation studies and runtime

analysis, demonstrate the effectiveness of the integrated pipeline for query-driven object-relationship detection.

The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 describes the proposed methodology, Section 4 presents experimental results and discussion, and Section 5 concludes the paper with future directions.

2 Related Work

Our research related to our study can be grouped into three areas: Open-Vocabulary Object Detection (OVOD), Scene Graph Generation (SGG), and Vision–Language Models (VLMs). Prior research relevant to this study spans Open-Vocabulary Object Detection (OVOD), Scene Graph Generation (SGG), and vision–language integration. Traditional detectors such as Faster R-CNN [10] and YOLO [22] operate in closed-vocabulary settings, motivating OVOD approaches that leverage multimodal representations from CLIP [15], including ViLD [16] and OWL-ViT [17], which enables detection from arbitrary text queries. Beyond object-level understanding, SGG models describe structured relationships (e.g., <person, riding, horse>) using Transformer-based methods such as RelTR [1] and SGTR [18], commonly trained on datasets like Visual Genome [19]. For language grounding, sentence embedding–based vision–language approaches such as MDETR [21] align text phrases with image regions but do not explicitly construct full scene graphs, motivating the integrated pipeline proposed in this work.

3 Proposed Method

OWL-ViT (Open-World Localization Vision Transformer) [17] is adopted as the primary detection module due to its open-vocabulary capability. Extending the CLIP [15] framework, OWL-ViT integrates a Vision Transformer image encoder with a text encoder to map image regions and textual queries into a shared embedding space, enabling detection of previously unseen categories without retraining. Compared with approaches such as ViLD [16], OWL-ViT supports flexible query-driven detection while maintaining strong generalization. For relationship reasoning, **RelTR (Relation Transformer)**, an extension of DETR [7], is employed to identify structured interactions between detected objects. RelTR extracts global contextual features through a Transformer encoder and directly predicts subject–relation–object triplets using learned triplet queries, optimized via Hungarian matching. This design allows the model to learn relational representations without rule-based constraints and generalize to unseen relationships. For query understanding, **Sentence Transformers** are used to encode natural language input into semantic embeddings. While vision–language models such as MDETR [21] can ground text phrases to image regions, they do not explicitly generate structured scene graphs. Therefore, the proposed pipeline combines OWL-ViT for detection, RelTR for relationship inference, and sentence embeddings for query processing to enable efficient query-driven object–relationship analysis.

3.1 Introduction to the Proposed Model

The system's workflow involves receiving user queries, detecting objects in the image, identifying relationships between them, and extracting corresponding bounding boxes to display the results. This system enables users to search for visual information more intuitively and naturally compared to traditional methods. The detected results are standardized in coordinates for consistent data representation. Next, the RelTR model identifies relationships between the objects, while the SENs analyzes the query to extract keywords and intent. The system then compares these relationships with the query keywords to filter out only the most relevant information and finally outputs the image with highlighted objects and relationships corresponding to the query.

Data Processing: In the integrated OWL-ViT–RelTR framework, images are filtered, resized to compatible resolutions, and normalized. OWL-ViT processes natural language queries to detect objects and produce bounding boxes, which are converted for RelTR. RelTR infers subject–relation–object triplets, which are then interpreted by the language module to return query-relevant results, enabling coherent object localization, relationship reasoning, and query-driven visual analysis.

3.2 Training Process

The proposed multi-method pipeline integrates open-vocabulary object detection, relationship reasoning, and query-driven filtering into a unified system. The process begins with training OWL-ViT on Visual Genome, divided into training and validation sets. Training is performed using GPU resources for 100 epochs, resulting in an object detection model with approximately 85% accuracy. After object detection, the Relation Transformer (RelTR) is trained to infer semantic relationships between detected objects. OWL-ViT provides bounding boxes, labels, and feature representations, which are encoded and processed by RelTR to predict structured triplets (subject, relation, object) based on spatial and contextual cues. This enables the identification of interactions such as positional or functional relationships between objects. To improve detection flexibility, OWL-ViT supports open-vocabulary labeling, allowing previously unseen object categories to be recognized and integrated into the relationship analysis stage. The refined detection outputs are standardized and passed to RelTR for comprehensive scene understanding. Finally, a Sentence Transformer–based language module processes user queries by extracting semantic information and filtering predicted triplets to match the requested objects and relationships. The integrated system is then optimized to ensure stable performance and efficient deployment for query-driven visual analysis tasks.

Evaluation Metrics: The evaluation employs standard classification metrics, including Accuracy, Precision, Recall, and F1-score, and extends them with a generalized metric combining F1 and mAP@50 to assess overall system performance. Experiments are conducted on benchmark datasets (COCO, LVIS, Visual Genome, VRD) and additional real-world data. Predictions from each module are compared with unified ground truth annotations using confusion matrix definitions, and results are averaged across multiple runs.

3.3. Experiments Results

We evaluate the system on multiple datasets covering object detection and scene graph generation. Visual Genome (VG) contains approximately 108k images, 75k object categories, and 40k relation types, using a 70/10/20 train–validation–test split. The VRD dataset includes about 5k images, 100 object categories, and 70 predicates. For open-vocabulary detection, COCO and LVIS are used to assess localization and classification performance. To benchmark performance, we compare our proposed pipeline against several representative models. These include OWL-ViT [17] as the open-vocabulary object detection baseline, RelTR [1] as an end-to-end relation detection framework with its own detection module, and MDETR, a multimodal Transformer designed for text–image grounding. We also consider ViLD [16], which distills CLIP into an object detector for open-vocabulary tasks, and, as a classical SGG baseline, Faster R-CNN combined with MotifNet.

Results

During the testing of the OWL-ViT model in zero-shot mode, meaning it uses text prompts directly without retraining on the target dataset, the results showed relatively high accuracy.

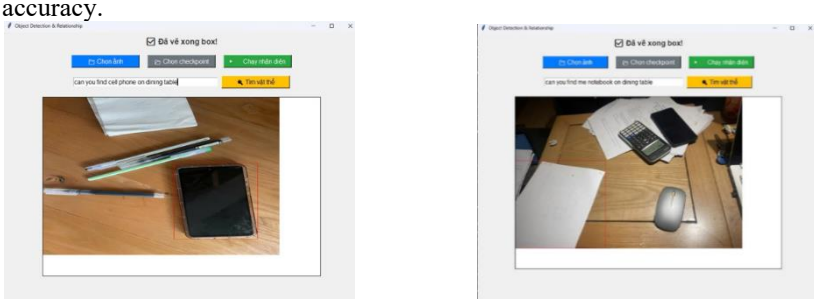


Fig 1. An image is fed into the model with the query "can you find a cellphone on dining table", and an image is fed into the model with another query "can you find me notebook on dining table". The results illustrated in Table 1 indicate that OWL-ViT provides a strong zero-shot baseline across datasets, while the proposed pipeline achieves the highest overall detection performance by combining open-vocabulary detection with relationship-aware filtering.

Table 1: Object Detection and OVOD Performance across Datasets and Models

Model / Scenario	Dataset	mAP@50 (%)	mAP@95 (%)	Recall (%)	Precision (%)	Notes
OWL-ViT	COCO (zero-shot)	34.5	19.0	40.1	29.0	Standard CLIP prompts, no fine-tuning
OWL-ViT	LVIS (long-tail)	30.2	–	35.6	26.3	Maintains performance despite rare classes
OWL-ViT	Real-world images	28–36	–	33–41	24–30	Performance varies due to occlusion/unseen objects
OWL-ViT (expanded prompts)	Average	35.0	–	38.0	27.0	Improved detection without fine-tuning
ViLD	COCO/LVIS	32.0	18.0	–	–	CLIP-based OVOD baseline
RelTR (built-in detector)	COCO/LVIS	25.0	13.8	–	–	Lower detection accuracy

MDETR	COCO/LVIS	28.0	15.3	–	–	Strong grounding, weaker SGG
Faster R-CNN + MotifNet	COCO/LVIS	22.0	11.0	–	–	Closed-vocabulary baseline
Proposed Pipeline	COCO/LVIS	**36.0**	**22.6**	–	–	OWL-ViT + RelTR + language filtering

Also, RelTR is the most efficient (120 ms, 6 GB, 200 M params), while OWL-ViT requires 150 ms and 8 GB. MDETR is heavier (200 ms, 10 GB, 350 M). The proposed pipeline is the most resource-intensive (250 ms, 12 GB, 500 M) due to integrating multiple modules for improved accuracy and flexibility. The results in Table 2 are derived from the OWL-ViT paper, along with practical tests from the GitHub OWL-ViT demo and benchmarks on COCO & LVIS, as well as internal results from several projects implementing the OWL-ViT zero-shot model on medical/agricultural images. The object recognition test results indicate that OWL-ViT's practical recognition capability is 35%, which is relatively good.

Table 2. Relation, Detection, and Query Performance with Ablation (Visual Genome)

Model / Configuration	**mAP @ 50**	**R@ 20**	**R@ 50**	**R@ 100**	**Relation R@50**	**P**	**R**	**F1**	**Fβ,γ**	**Notes**
OWL-ViT (only)	34.5	12.5	18.0	22.0	N/A	35.0	20.0	25.4	21.0	Strong detection, no relation reasoning
RelTR (only)	25.0	25.0	32.0	38.0	32.0	30.0	33.0	31.4	29.0	Relation inference, weak detection
OWL-ViT → RelTR	34.0	32.0	40.0	46.0	40.0	38.0	36.0	37.0	34.5	Relation inference without query filtering
MDETR	–	28.0	36.0	42.0	–	34.0	34.0	34.0	31.0	Strong grounding, limited SGG
Faster R-CNN + Motifs	–	20.0	28.0	34.0	–	26.0	27.0	26.5	24.0	Closed-vocabulary baseline
Proposed Pipeline (Full)	36.0	38.0	47.0	52.0	47.0	42.0	41.0	41.5	39.0	Best overall performance

3.3.3 Conclusion

This work presents a unified integration of OVOD, SGG, and semantic query filtering to address query-based visual relationship detection. Unlike single-model approaches, the proposed system jointly models object relationships and supports natural language query interpretation. By combining open-vocabulary detection with structured relationship reasoning, the pipeline offers greater flexibility than MDETR, OWL-ViT, or RelTR used independently. Experimental results show improved performance in both object localization and relation inference, better satisfying user queries. Future work will focus on hyperparameter optimization, expanding training data, and integrating additional models such as Region-CLIP to further enhance recognition of novel object categories.

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